

Measured Layout Preservation in Document Translation

A. Verma, L. Okafor, M. Reinhold · Institute for Document Systems

Abstract

We report a controlled evaluation of layout-preserving document translation across five target languages. Our method reconstructs each text block in place, measured against a pixel-level fidelity criterion, and reports the typographic concession made whenever translated text does not fit the original frame. On a corpus of 412 synthetic pages the approach preserves graphics fidelity at 0.994 masked SSIM while leaving 2.1% of blocks flagged for human review [3], [7].

1 Introduction

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Our contribution is narrower and, we argue, more useful: a reconstruction loop that erases only the text layer of a block, then re-replaces the translated string in the same rectangle under an explicit ladder of typographic concessions, recording which rung was required. The ladder is ordered by perceptual cost, so the cheapest change that fits is the change that is made [4], [5].

Section 3 defines the fit ladder and its bounds. Section 4 describes the protection scheme that keeps numbers, identifiers and citations verbatim through the translation step. Section 5 reports measured fidelity on the evaluation corpus, and Section 6 discusses the failure modes we could not eliminate, chiefly font substitution on branded material.

The evaluation corpus consists of synthetic pages generated with known ground truth, which lets us separate parsing error from translation error. Each page records its authored element inventory, so a parse can be scored directly rather than judged by inspection [6].

Table 1 reports fidelity by target language. Contraction (Chinese, Japanese) is uniformly easier than expansion (German, French): a shorter string always fits, so the ladder is never entered, and the residual error is confined to line-breaking quality inside the original frame.

Expansion behaves differently. The mean concession rises with the expansion ratio, and the tail matters more than the mean: a single clipped clause in a legal document is worse than a page set 4% smaller. We therefore report the worst rung reached per page alongside the mean.

Threats to validity: synthetic pages under-represent the designed layouts found in magazines and marketing material, where text frames are irregular and often overlap artwork. We expect materially worse reflow there and say so rather than extrapolating from the corpus we measured.

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3 The fit ladder

$$\text{cost}(r) = w_r \cdot \text{area}(b) / \text{area}(\text{page}), \quad w = (0,1,2,3,5,8,13)$$

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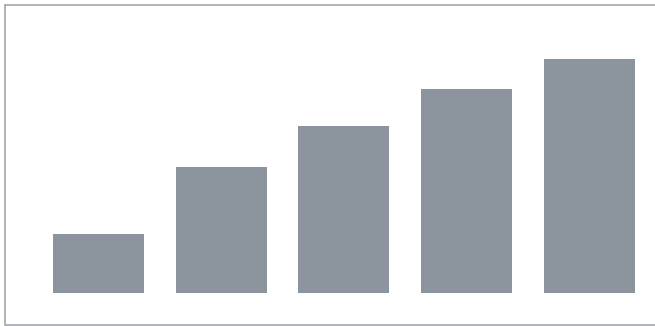


Figure 1: Mean concession by target language, mock provider, $n = 412$ pages.

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Target	Ratio	SSIM	Rung>3	Flag %
German	1.32	0.993	38	2.9
French	1.22	0.994	21	1.8
Hindi	1.18	0.995	17	1.5
Japanese	0.68	0.997	2	0.4

Table 1: Fidelity by target language. Ratio is mean character expansion.

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1. Ground truth for the corpus is published with the generator script.
2. Ratios measured on in-domain prose, not headlines.